deal with Japan, this region remained cut-off; it took about a week to reach the area from its nearest railhead in Raipur or Visakhapatnam.

Boosting business

The new agreement led to the laying of three new railway lines, which connected the region and ensured a seamless transfer of iron ore from the mines to the Visakhapatnam port in Andhra Pradesh.

The 41-km Kiriburu–Bimlagarh link and the 182-km Sambalpur–Titlagarh line transported ore from the Kiriburu mines. The 448-km-long Kothavalasa–Kirandul line connected the Bailadila mines at Kirandul with the South Eastern Railway's main line at Kothavalasa, 27 km from Waltair, the port head of Visakhapatnam and a key freight transit point.

The challenges

The first two lines, Kiriburu–Bimlagarh and Sambalpur–Titlagarh, both in Odisha, opened for use in April 1963. Kothavalasa–Kirandul, also known as the K–K line, proved to be an engineering challenge.

Passing over the Eastern Ghats, the line has the distinction of being the highest broad-gauge line in the world as it gains an altitude of 997 m above sea level. The difficult terrain in the Ghat section necessitated gradients of 1 in 60 and curves as sharp as eight degrees (radius 218.38 m). The construction of the line involved boring 46 tunnels and building 87 major and 1,236 minor bridges.

Tunnels proved to be particularly challenging. The oldest hill range in India, the Eastern Ghats are formed from a foliated metamorphic rock called khondalite. So, work on them required high technical competence as the boring involved cutting through broken, stratified, and weathered strata. Invariably, the top heading had to be constructed and supported by steel ribs with the benching through underpinning operations.

Chimneys, domes, and vaults occurred in some of the tunnels, reducing access and making it difficult for workers to carry heavy machinery. Regardless of the challenges, work carried on manually. Engineers had to battle frequent landslides and steep hillsides, which often resulted in the construction of bridges on steep, eight-degree curves. This meant that the erection of girders on the alignment became a problem. In some cases, gorges were bridged using 40.26-m-high